

# Q/SZQY Growatt New Energy Co., Ltd. Enterprise Standard

Q/SZQY—20002—2018

**Growatt BMS CAN-Bus-protocol -low-voltage**

| Rev.           | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Author   |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| V1.01 2018/6/1 | Our company's BMS protocol Protocol_CAN_20180601                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Wei Wei  |
| V1.02          | <p>2018/11/13:</p> <ol style="list-style-type: none"> <li>CAN ID: 0x312 Increase battery cell count;</li> <li>CANID: 0x315~0x318 Increase the voltage of individual cells 1~16;</li> <li>Added battery model 6532 to CAN ID:0x320;</li> </ol> <p>2018/12/6:</p> <ol style="list-style-type: none"> <li>CANID: 0x319 Add battery type definition Byte0 bit0-bit1 (00: Phosphate)<br/>Lithium iron phosphate battery / 01: Ternary lithium battery / 10: Lithium titanate battery / 11: Reserved);</li> <li>CAN ID: 0x320 (remove battery model 6532, Bytes 6-7 are used for Date &amp; Time)</li> </ol> | Wang Min |
| V1.03          | <p>2019/02/19:</p> <ol style="list-style-type: none"> <li>CAN ID: 0x0319 Add reporting of the highest and lowest voltage of a single unit.</li> <li>Add CAN port definition description for the energy storage unit.</li> <li>CANID: 0x315~0x318 is defined as optional; the battery can choose to report it.</li> <li>CAN ID: 0x0319 Battery ID used for reporting fault protection when adding parallel connections.</li> </ol>                                                                                                                                                                      | Demon    |
| V1.04          | <p>2019/02/22:</p> <ol style="list-style-type: none"> <li>CAN ID: 0x321 Add Remote Upgrade Reporting (ATL) information.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Demon    |
| V1.05          | <p>2019/08/28:</p> <p>Add a write data ID definition, CAN ID: 0x211. Add frequency modulation enable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Demon    |
| V1.06          | <p>2020/03/23</p> <p>0x322~0x325 Add fault and alarm information and battery list reported in parallel with the battery<br/>Body information, etc.</p> <p>0x211 Add battery time synchronization information</p> <p>0x312 Reason for power reduction after adding battery</p> <p>Information not applicable in other optimization software</p>                                                                                                                                                                                                                                                         | Demon    |
| V1.07          | <p>2020/04/30</p> <p>0x320 Add battery reporting software version high byte extension</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Demon    |
| V1.08          | <p>2020/06/19</p> <p>0x211 is defined as an event-driven instruction, enabling battery fault clearing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Demon    |

## 1. CAN communication method

### ȳ CAN Bus Specifications

Standard frames are used, and the bus transmission rate is 500kbps.

### ȳ Data Pattern

Big-endian, where the high-order byte of data is stored at the address of the low-order byte (example: Data = byte3 \* 256 + byte4). This will be used in the protocol.

The following data types are defined as follows:

| Sequence | Number | Data Type Definition | Data length (bytes) |
|----------|--------|----------------------|---------------------|
| 1        | Byte   | unsigned char        | 1                   |
| 2        | Uint16 | unsigned short int   | 2                   |
| 3        | Uint32 | unsigned int         | 4                   |
| 4        | Sint8  | signed char          | 1                   |
| 5        | Sint16 | signed short int     | 2                   |
| 6        | Sint32 | signed int           | 4                   |
| 7        | FP32   | float                | 4                   |

### Communication Mode

After the energy storage device sends a query command or control command frame, the battery pack device responds with data;

Heartbeat frame: The inverter sends data every second.

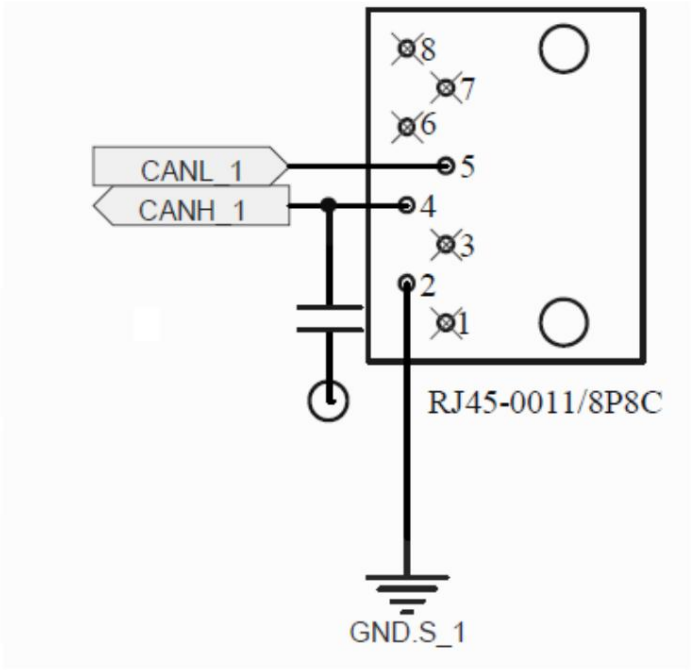
### CAN ID:0x301

|        |      |             |             |
|--------|------|-------------|-------------|
| Byte 0 | NULL | <b>0x0B</b> | Fixed value |
| Byte 1 | NULL | 0x16        | Fixed value |
| Byte 2 | NULL | <b>0x21</b> | Fixed value |
| Byte 3 | NULL | 0x2C        | Fixed value |
| Byte4  | NULL | <b>0x37</b> | Fixed value |
| Byte 5 | NULL | 0x42        | Fixed value |
| Byte 6 | NULL | 0x4d        | Fixed value |
| Byte 7 | NULL | 0x58        | Fixed value |

This data frame is sent by the energy storage device and is used to perform compatibility identification and communication connection determination for the battery; the battery should read this frame.

Data or perform an empty mailbox operation to prevent mailbox overflow;

### Interface Definition



Reading data frame format

| StdId              | ExtId | GO | RTR | DLC | Data[8]   | FMI |
|--------------------|-------|----|-----|-----|-----------|-----|
| ID area<br>y0x3xxÿ | NC    | NC | NC  | NC  | DATA area |     |

Write data frame format

| StdId                    | ExtId | GO | RTR | DLC | Data[8]   | FMI |
|--------------------------|-------|----|-----|-----|-----------|-----|
| ID area<br>(0x4xx\0x2xx) | NC    | NC | NC  | NC  | DATA area |     |

ÿ Glossary

SP: Energy Storage Unit

Cell: Battery cell

Pack: A battery pack that encapsulates the BMS system and typically contains multiple battery cells.

FCC: Battery full load capacity

RM: Remaining capacity

## 2. CAN Message - Write Data

### CAN ID: 0x211

|        |                                                     |        |                                    |
|--------|-----------------------------------------------------|--------|------------------------------------|
| Byte 0 | frequency adjustment<br>Enable frequency modulation |        | 1yEnable<br>0yDisable<br>Defaulty0 |
| Byte 1 | Year                                                | 20~250 | Battery time synchronization       |
| Byte 2 | Month                                               | 1~12   |                                    |
| Byte 3 | Day                                                 | 1~31   |                                    |
| Byte4  | hour                                                | 0~23   |                                    |
| Byte 5 | Minute                                              | 0~59   |                                    |
| Byte 6 | Second                                              | 0~59   |                                    |
| Byte 7 | Troubleshooting                                     | 0~1    | 1yEnable<br>0yDisable<br>Defaulty0 |

Event-based data commands are triggered by an event, with a 1-second sending cycle, sending 5 times consecutively, and time synchronization triggered on the hour.

### CAN ID: 0x212

|         |                                         |  |                          |
|---------|-----------------------------------------|--|--------------------------|
| Byte 0  | Battery parameter query enabled. Comand |  |                          |
| Byte 1  | Table: Battery ID                       |  | Battery ID to be queried |
| Byte 2  | (Reserved).                             |  |                          |
| Byte 3  | Reserved                                |  |                          |
| Byte4   | Reserved                                |  |                          |
| Byte 5  | Reserved                                |  |                          |
| Byte 6  | Reserved                                |  |                          |
| Byte 7: | Reserved                                |  |                          |

This data frame is used to read specified information from the battery when needed. It does not need to be sent periodically; the battery sends the information after receiving it.

One request, one response;

Command Table: Battery Special Parameter Query Command

|                                                       |                                      |       |     |
|-------------------------------------------------------|--------------------------------------|-------|-----|
| SNGetEn                                               | Battery serial number lookup enabled | BYTE1 | 0.0 |
| HistoryFaultGetEN Enables Battery History Fault Query |                                      | BYTE1 | 0.1 |
| Res                                                   | Reserved                             | BYTE1 | 0.2 |
| Res                                                   | Reserved                             | BYTE1 | 0.3 |
| Res                                                   | Reserved                             | BYTE1 | 0.4 |
| Res                                                   | Reserved                             | BYTE1 | 0.5 |
| Res                                                   | Reserved                             | BYTE1 | 0.6 |
| Res                                                   | Reserved                             | BYTE1 | 0.7 |

### 3. CAN Message - Read Data

#### CAN ID: 0x311

|        |                                                 |            |                        |
|--------|-------------------------------------------------|------------|------------------------|
| Byte 0 | Recommended charging voltage (CV) for batteries | Unit: 0.1V | Uint16, 2's complement |
| Byte 1 |                                                 |            |                        |
| Byte 2 | Charge current limit                            | Unit: 0.1A | Uint16, 2's complement |
| Byte 3 |                                                 |            |                        |
| Byte4  | Discharge current limit                         | Unit: 0.1A | Uint16, 2's complement |
| Byte 5 |                                                 |            |                        |
| Byte 6 | Status                                          | Bit0~11    | <b>Table1</b>          |
| Byte 7 |                                                 |            |                        |

**Table1: Status bits**

| Bit Index | Content                                                                           | Comment                                            |
|-----------|-----------------------------------------------------------------------------------|----------------------------------------------------|
| 0         | status                                                                            | 00 : soft_starting                                 |
| 1         |                                                                                   | 01 : stand by<br>10 : charging<br>11 : discharging |
| 2         | Error bit flag                                                                    | 1 : "Error" byte valid<br>0 : "Error" byte Invalid |
| 3         | Cell balance status                                                               | 0 : unbalance<br>1 : balance                       |
| 4         | Sleep status                                                                      | 0 : disable<br>1 : enable                          |
| 5         | Output Discharge status                                                           | 0 : disable<br>1 : enable                          |
| 6         | Output Charge status                                                              | 0 : disable<br>1 : enable                          |
| 7         | Battery terminal status                                                           | 0 : terminal connected<br>1 : terminal open        |
| 8         | Master box Operation Mode: 00: Standalone; 01: Parallel; 10: Parallel preparation |                                                    |
| 9         |                                                                                   |                                                    |
| 10        | Reserved                                                                          |                                                    |
| 11        |                                                                                   |                                                    |

"Master box Operation Mode" There is no special control in the current SP program ,

All

The controls are performed by the BMS itself; no special . SP only identifies the state. (This is in the context of current energy storage technology.) controls are implemented. All controls are handled by the battery BMS itself, and the energy storage device is only used to identify the state.

#### CAN ID: 0x312

|        |                      |                |  |
|--------|----------------------|----------------|--|
| Byte 0 | Battery Protection 1 | <b>Table 1</b> |  |
|--------|----------------------|----------------|--|

|        |                                                                      |                |       |
|--------|----------------------------------------------------------------------|----------------|-------|
| Byte 1 | Battery Protection                                                   | <b>Table 2</b> |       |
| Byte 2 | 2 Battery Alarm                                                      | <b>Table 3</b> |       |
| Byte 3 | 1 Battery Alarm 2                                                    | <b>Table 4</b> |       |
| Byte4  | <b>Pack Number</b><br><small>Number of batteries in parallel</small> | 1~254          | Uint8 |
| Byte5  | Reasons for battery power reduction (H)                              | <b>Table 5</b> |       |
| Byte6  | Reasons for battery power reduction (L)                              | <b>Table 5</b> |       |
| Byte 7 | Reserved                                                             |                |       |

**Table 1**

| Bit 7                  | Bit 6               | Bit 5                                   | Bit 4             | Bit 3              | Bit 2               | Bit 1                | Bit 0           |
|------------------------|---------------------|-----------------------------------------|-------------------|--------------------|---------------------|----------------------|-----------------|
| DisCharge over current | Charge over current | SCD(Short Circuit Discharge) protection | Cell over voltage | Cell under voltage | module over voltage | module under voltage | Soft start fail |

**Table 2**

| Bit 7                                      | Bit 6                                   | Bit 5                                       | Bit 4                                    | Bit 3         | Bit 2        | Bit 1            | Bit 0                         |
|--------------------------------------------|-----------------------------------------|---------------------------------------------|------------------------------------------|---------------|--------------|------------------|-------------------------------|
| OTD(Over Temperature Discharge) protection | OTC (Over Temperature Charge)protection | UTD (Under Temperature Discharge)protection | UTC (Under Temperature Charge)protection | Syste m error | Delta V Fail | in parallel fail | Parallel version Inconsistent |

**Table 3**

| Bit 7                  | Bit 6               | Bit 5 | Bit 4             | Bit 3              | Bit 2               | Bit 1                | Bit 0 |
|------------------------|---------------------|-------|-------------------|--------------------|---------------------|----------------------|-------|
| DisCharge over current | Charge over current |       | Cell over voltage | Cell under voltage | module over voltage | module under voltage |       |

**Table 4**

| Bit 7                                       | Bit 6                                     | Bit 5                                         | Bit 4                                      | Bit 3 | Bit 2        | Bit 1                             | Bit 0                       |
|---------------------------------------------|-------------------------------------------|-----------------------------------------------|--------------------------------------------|-------|--------------|-----------------------------------|-----------------------------|
| OTD(Over Temperature and Discharge) warning | OTC (Over Temperature and Charge) warning | UTD (Under Temperature and Discharge) warning | UTC (Under Temperature and Charge) warning |       | Delta V Fail | Battery Power off Previous police | Internal communication fail |

**Table 5**

|               |                                       |       |     |
|---------------|---------------------------------------|-------|-----|
| HwFltLimiFlg  | Hardware failure current limiting and | BYTE1 | 5.0 |
| ChFullLimiFlg | full charge current limiting          | BYTE1 | 5.1 |

|                  |                                                                                   |       |     |
|------------------|-----------------------------------------------------------------------------------|-------|-----|
| TmosHLimiFlg     | MOS overheating current limiting, ambient                                         | BYTE1 | 5.2 |
| TenvHLimiFlg     | temperature overheating current limiting,                                         | BYTE1 | 5.3 |
| PchFltLimiFlg    | precharge fault current limiting,                                                 | BYTE1 | 5.4 |
| ComFltLimiFlg    | communication fault current limiting,                                             | BYTE1 | 5.5 |
| CBusFltLimiFlg   | bus fault current limiting, reserved                                              | BYTE1 | 5.6 |
| Res              | cell                                                                              | BYTE1 | 5.7 |
| UcellHLimiFlg    | overheating current limiting, cell                                                | BYTE0 | 6.0 |
| UcellLLimiFlg    | underheating current limiting,                                                    | BYTE0 | 6.1 |
| TcellHLimiFlg    | temperature overheating current                                                   | BYTE0 | 6.2 |
| TcellLLimiFlg    | limiting, temperature underheating                                                | BYTE0 | 6.3 |
| UmainHLimiFlg    | current limiting, total voltage                                                   | BYTE0 | 6.4 |
| UmainLLimiFlg    | overheating current limiting, total                                               | BYTE0 | 6.5 |
| UcellDiffLimiFlg | voltage underheating current limiting,                                            | BYTE0 | 6.6 |
| TDiffLimiFlg     | cell voltage difference current limiting, temperature difference current limiting | BYTE0 | 6.7 |

**CAN ID: 0x313**

|        |                                                                                                                     |             |                                                           |
|--------|---------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------|
| Byte 0 | Voltage of single module or average module voltage of The voltage of a single module or the system. Average voltage | Unit: 0.01V | Sint16, 2's complement                                    |
| Byte 1 |                                                                                                                     |             |                                                           |
| Byte 2 | Module or system total current Single unit or total system current                                                  | Unit: 0.1A  | Sint16, 2's complement                                    |
| Byte 3 |                                                                                                                     |             |                                                           |
| Byte4  | Battery average temperature                                                                                         | Unit: 0.1y  | Sint16, 2's complement                                    |
| Byte 5 |                                                                                                                     |             |                                                           |
| Byte 6 | Average SOC                                                                                                         | Unit: 1%    | Uint8                                                     |
| Byte 7 | SOH                                                                                                                 |             | Bit 0~ Bit6 SOH<br>Bit7: Battery Life Warning Sign<br>Zhi |

**CAN ID: 0x314**

|        |             |       |                                                     |
|--------|-------------|-------|-----------------------------------------------------|
| Byte 0 | Gauge RM    | 10mAh | Current capacity                                    |
| Byte 1 |             |       |                                                     |
| Byte 2 | Gauge FCC   | 10mAh | Normal fully charged capacity                       |
| Byte 3 |             |       |                                                     |
| Byte4  | Delta V     | 1mV   | Difference between the max Cell voltage and the min |
| Byte 5 |             |       |                                                     |
| Byte 6 | Cycle Count |       |                                                     |
| Byte 7 |             |       |                                                     |



**CAN ID: 0x319**

| Byte 0 | Request& battery type                     | Table 6 |        |
|--------|-------------------------------------------|---------|--------|
| Byte 1 | Maximum cell voltage                      | 1mV     | Uint16 |
| Byte 2 |                                           |         |        |
| Byte 3 | Minimum cell voltage                      | 1mV     | Uint16 |
| Byte4  |                                           |         |        |
| Byte 5 | Maximum cell voltage number               | 1       | Uint8  |
| Byte6  | Minimum cell voltage number               | 1       | Uint8  |
| Byte7  | Protect pack ID<br>Faulty battery address | 1       | Uint8  |

**Note : When the batteries are connect in parallel ,Cell number starts with the mainfram ,and then the slave .When reporting the highest or lowest voltage of a cell , all the cells should be counted.**

Note: When batteries are connected in parallel, the battery cell numbering starts from the master unit, then proceeds sequentially to the first slave unit, second slave unit, etc., and the parallel connection is reported.

The highest or lowest voltage for a single cell is calculated by statistically analyzing all cells together.

**Table 6**

| Bit 7         | Bit 6            | Bit 5                                              | Bit 4                                               | Bit 3 | Bit 2 | Bit 1                                                                                                                  | Bit 0 |
|---------------|------------------|----------------------------------------------------|-----------------------------------------------------|-------|-------|------------------------------------------------------------------------------------------------------------------------|-------|
| Charge enable | Discharge enable | Request force<br>charge I*<br>strong charge<br>ÿÿ1 | Request force<br>Charge II*<br>Strong Charge Mark 2 |       |       | 00: Lithium iron phosphate battery<br>01: Ternary lithium battery<br>10: Lithium titanate battery<br><b>11: Retain</b> |       |

Please use bit 5, the SOC range is: 5~10%. Bit 4 is NULL.

In this case, inverter itself should set a threshold of SOC: after force charge, only when battery SOC is higher than this threshold then inverter will allow discharge, to avoid force charge and discharge status change frequently.

**CAN IDÿ0x320**

This data frame is used to identify the battery manufacturer; batteries compatible with this protocol must contain this data frame. If the battery does not have a specific...

For special needs requiring compatibility processing by the energy storage device, the manufacturer's name abbreviation for this data frame can be 0; the time parameter is currently not used.

Energy storage devices have clock systems.

|                  |                                                                    |                       |                                   |
|------------------|--------------------------------------------------------------------|-----------------------|-----------------------------------|
| Byte 0           | Manufacturer Name                                                  | Growatt               | Example:Byte 0<br>="G",Byte 1="T" |
| Byte 1           |                                                                    | ASCII                 |                                   |
| Byte 2           | Hardware version                                                   | Low hardware          | rangeÿ1~9                         |
| Byte 3           | Software version L                                                 | and software versions | rangeÿ1~9                         |
| Byte 4           | Software version H (high byte extension)                           |                       |                                   |
| Byte 5's highest | concurrent software version and lowest concurrent software version |                       |                                   |

|                                                                                            |  |  |
|--------------------------------------------------------------------------------------------|--|--|
| Byte 6, highest concurrent software version, maximum concurrent software version extension |  |  |
| Byte 7 reserved                                                                            |  |  |

\*1 Note: Manufacturer Name (uppercase letters of the battery manufacturer's abbreviation);

\*2 For the high byte of the software version extension, if the battery version has only one byte, only the low byte is reported, and the high byte is reported as 0.

#### CAN ID 0x321

|                 |                                                                            |         |             |
|-----------------|----------------------------------------------------------------------------|---------|-------------|
| Byte 0          | Update status<br>Upgrade status                                            | Table 6 |             |
| Byte 1          | <del>Update schedule of pack</del><br><del>Battery upgrade progress</del>  |         | range 0~100 |
| Byte 2          | <del>programming Status of pack</del><br><del>Battery upgrade status</del> |         |             |
| Byte 3 reserved |                                                                            |         |             |
| Byte 4 reserved |                                                                            |         |             |
| Byte 5 reserved |                                                                            |         |             |
| Byte 6 reserved |                                                                            |         |             |

The battery supports remote upgrades, and the upgrade process is reported to the energy storage unit. If remote upgrades are not supported, this data frame is not used. Upgrade successful.

Issued after restart. (Non-periodic command)

**Table 6**

| Bit 5~Bit 7 Bit 3~Bit 4                                                                                                                       | <del>Bit 1 ~ Bit 2</del>                                                                                                                          | Bit 0~1                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <del>00: Slave normal</del><br><del>01: Slave programming</del><br><del>10: Slave update successful</del><br><del>11: Slave update fail</del> | <del>00: Master normal</del><br><del>01: Master programming</del><br><del>10: Master update successful</del><br><del>11: Master update fail</del> | 0: normal<br>(operating normally)<br>1: programming<br>Upgrading<br>2: Upgrade successful |

#### CAN ID 0x322

|                                        |                      |        |
|----------------------------------------|----------------------|--------|
| Byte 0 highest unit temperature        | 0.1℃                 | Sint16 |
| Byte 1                                 | Byte 0 *256 + Byte 1 |        |
| Byte 2 Minimum Temperature             | 0.1℃                 | Sint16 |
| Byte 3                                 |                      |        |
| Byte 4 Highest Temperature Unit Number |                      | Byte   |
| Byte 5 Lowest Temperature Unit Number  |                      | Byte   |
| Byte 6 Highest SOC                     | 0~100                | Uint8  |
| Byte 7 Minimum SOC                     | 0~100                | Uint8  |

**CAN ID 0x323**

|                             |                                                                   |       |       |
|-----------------------------|-------------------------------------------------------------------|-------|-------|
| Byte 0                      | Total Cell Number<br><small>Total number of battery cells</small> | 1~254 | Uint8 |
| Byte 1 reserved             |                                                                   |       |       |
| Byte 2 reserved             |                                                                   |       |       |
| Byte 3 reserved             |                                                                   |       |       |
| Byte 4 Battery Protection 3 |                                                                   |       |       |
| Byte 5 Battery Protection 4 |                                                                   |       |       |
| Byte 6 Battery Protection 5 |                                                                   |       |       |
| Byte 7 Battery Warning 3    |                                                                   |       |       |

|                                                                                        |                                                                          |              |     |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------|-----|
| Battery alarm 3                                                                        |                                                                          |              |     |
| OLC                                                                                    | Overcharging power                                                       | Byte 7       | 7.0 |
| OLD                                                                                    | Over Power Discharging                                                   | Byte 7       | 7.1 |
| FLT_PRLL_I_INCH_H2 System internal charging circulating current overcurrent warning    |                                                                          | Byte 7       | 7.2 |
| FLT_PRLL_I_INDISE_H2 System internal discharge circulating current overcurrent warning |                                                                          | Byte 7       | 7.3 |
| Battery protection 3                                                                   |                                                                          |              |     |
| OLC                                                                                    | Over Power Charging                                                      | Byte 4       | 4.0 |
| OLD                                                                                    | Over-power Discharging External                                          | Byte 4       | 4.1 |
| EXT_COM                                                                                | Communication Failure                                                    | Byte 4       | 4.2 |
| Pre_Chg_Fail                                                                           | PreCharge Fail                                                           | Byte 4       | 4.3 |
| HW_Fault                                                                               | BMS HW Fault                                                             | Byte 4       | 4.4 |
| AFE_COM                                                                                | Communication error<br>between AFE and MCU                               | Byte 4       | 4.5 |
| FLT_CELL_LOST                                                                          | Individual cell abnormality (individual cell failure)                    | Byte 4 total | 4.6 |
| FLT_SP_IMAIN                                                                           | current sampling failure                                                 | Byte 4       | 4.7 |
| Battery Protection 4                                                                   |                                                                          |              |     |
| FLT_SP_UMAIN                                                                           | Total pressure sampling failure                                          | Byte 5       | 5.0 |
| FLT_SP_ULOAD                                                                           | load-side total voltage sampling failure;                                | Byte 5       | 5.1 |
| FLT_EEP_PARAM                                                                          | calibration parameter loading failure.                                   | Byte 5 bus   | 5.2 |
| FLT_CHBUS_RVRSE                                                                        | reverse connection                                                       | Byte 5       | 5.3 |
| FLT_OVP                                                                                | Hardware Overvoltage (Hardware Protection Feedback)                      | Byte 5       | 5.4 |
| FLT_OCP                                                                                | Hardware Overcurrent (Hardware Protection Feedback)                      | Byte 5       | 5.5 |
| FLT_PARALLEL                                                                           | Parallel Operation Failure                                               | Byte 5       | 5.6 |
| FLT_PRLL_UDIFF_OVER                                                                    | Large voltage difference from host battery (Byte 5) Battery protection 5 |              | 5.7 |
|                                                                                        |                                                                          |              |     |
| FLT_DIS_OCP                                                                            | Hardware discharge overcurrent fault (hardware protection)<br>(Feedback) | Byte 6       | 6.0 |
| FLT_CH_LIMIT_NORSP                                                                     | Charging current limiting failure fault,                                 | Byte 6       | 6.1 |
| FLT_UNLIMITED_NORSP                                                                    | discharging current limiting failure                                     | Byte 6       | 6.2 |
| FLT_BUS_OPEN                                                                           | fault, main circuit open circuit fault                                   | Byte 6       | 6.3 |



4. Respond to event trigger instructions

CAN ID 0x324

Return battery serial number

|                |               |        |        |        |        |        |        |
|----------------|---------------|--------|--------|--------|--------|--------|--------|
| Byte0          | Byte 1        | Byte 2 | Byte 3 | Byte 4 | Byte 5 | Byte 6 | Byte 7 |
| FrameNo<br>y0y | Battery<br>ID | SN0    | SN1    | SN2    | SN3    | SN4    | SN5    |

|                |        |        |        |        |        |        |        |
|----------------|--------|--------|--------|--------|--------|--------|--------|
| Byte0          | Byte 1 | Byte 2 | Byte 3 | Byte 4 | Byte 5 | Byte 6 | Byte 7 |
| FrameNo<br>y1y | SN6    | SN7    | SN8    | SN9    | SN10   | SN11   | SN12   |

|                |        |        |               |        |        |                   |        |
|----------------|--------|--------|---------------|--------|--------|-------------------|--------|
| Byte0          | Byte 1 | Byte 2 | Byte 3        | Byte 4 | Byte 5 | Byte 6            | Byte 7 |
| FrameNo<br>y2y | SN13   | SN14   | SN15 Reserved |        |        | Reserved Reserved |        |

Notice: If the battery serial number (SN) exceeds 16 bits, it can be extended according to the message format, but the maximum extension should not exceed 32 bytes.

CAN ID 0x325

Return to battery history faults (pending)

|                |               |        |        |        |        |        |        |
|----------------|---------------|--------|--------|--------|--------|--------|--------|
| Byte0          | Byte 1        | Byte 2 | Byte 3 | Byte 4 | Byte 5 | Byte 6 | Byte 7 |
| FrameNo<br>y0y | Battery<br>ID |        |        |        |        |        |        |

Detailed information on individual unit voltages will not be emphasized. For compatibility with older program versions, the CAN ID number will be retained, and this will be of highest priority in the program.

and lowest single-cell voltage;

CAN ID: 0x315

|        |                |     |        |
|--------|----------------|-----|--------|
| Byte 0 | Cell 1 Voltage | 1mV | Uint16 |
| Byte 1 |                |     |        |
| Byte 2 | Cell 2 Voltage | 1mV | Uint16 |
| Byte 3 |                |     |        |
| Byte 4 | Cell 3 Voltage | 1mV | Uint16 |
| Byte 5 |                |     |        |
| Byte 6 | Cell 4 Voltage | 1mV | Uint16 |
| Byte 7 |                |     |        |

CAN ID: 0x316

|        |                |     |        |
|--------|----------------|-----|--------|
| Byte 0 | Cell 5 Voltage | 1mV | Uint16 |
| Byte 1 |                |     |        |
| Byte 2 | Cell 6 Voltage | 1mV | Uint16 |
| Byte 3 |                |     |        |
| Byte 4 | Cell 7 Voltage | 1mV | Uint16 |

|                   |                           |                |                   |
|-------------------|---------------------------|----------------|-------------------|
| <del>Byte 5</del> |                           |                |                   |
| <del>Byte 6</del> | <del>Cell 8 Voltage</del> | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 7</del> |                           |                |                   |

**CAN ID:0x317**

|                   |                            |                |                   |
|-------------------|----------------------------|----------------|-------------------|
| <del>Byte 0</del> | <del>Cell 9 Voltage</del>  | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 1</del> |                            |                |                   |
| <del>Byte 2</del> | <del>Cell10 Voltage</del>  | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 3</del> |                            |                |                   |
| <del>Byte 4</del> | <del>Cell 11 Voltage</del> | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 5</del> |                            |                |                   |
| <del>Byte 6</del> | <del>Cell 12 Voltage</del> | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 7</del> |                            |                |                   |

**CAN ID:0x318**

|                   |                            |                |                   |
|-------------------|----------------------------|----------------|-------------------|
| <del>Byte 0</del> | <del>Cell 13 Voltage</del> | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 1</del> |                            |                |                   |
| <del>Byte 2</del> | <del>Cell14 Voltage</del>  | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 3</del> |                            |                |                   |
| <del>Byte 4</del> | <del>Cell 15 Voltage</del> | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 5</del> |                            |                |                   |
| <del>Byte 6</del> | <del>Cell 16 Voltage</del> | <del>1mV</del> | <del>Uint16</del> |
| <del>Byte 7</del> |                            |                |                   |